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Using Econometrics to Aid in Choosing Schools for My Sister

As (sister) is preparing to graduate from high school next spring, I understand that each of you has been contemplating the college application and admission process. I found data on all schools in the United States, and I would like to use my econometric expertise to help our family make good decisions when it comes to (sister) picking schools to apply to. I believe it is prudent to utilize the data we have at our disposal.

Section A – Sister

Sister – I understand that you want to mostly apply to schools you have a greater chance at getting into, and that your friends have suggested applying to more public schools than private schools. From the data I've provided, the average public college/university has an acceptance rate of 71.0%. However, the average private school actually has a higher acceptance rate, at 75.7% of applicants (4.7% higher than public schools), a point that my model is very confident about.

However, my model also thinks that whether a school is private, or public, only produces about 2.3% of the variation in acceptance rates across all schools. Therefore, I would urge you to apply to both public and private schools – there are lots of both that have both high and low acceptance rates. Just to get you thinking, here are some of the other things that have some effect on a college's acceptance rate:

1. For every 1,000 applications that a school receives, the acceptance rate lowers by 1.03%.
2. For every 100 point increase in the verbal SAT scores of newly enrolled students, there is a 9.1% drop in acceptance rate

This suggests that you should apply to schools that receive less applications, and tend to admit students with lower verbal SAT scores. The caveat with this, is that it may unfortunately be indicative of a poorer school or education. However, a regression including these statistics, and more, claimed to only account for 22.8% of variation in acceptance rate. If I were you, I would focus on finding a list of schools you are interested in, as there are countless factors playing into the acceptance rates of schools.

Section B – Mother

Mom, I realize that college is expensive, and that you are battling with the tradeoff between monetary cost and a high quality education. Therefore, I made a model that took a look at predicted out-of-state tuition, using the total SAT scores of current students, the percentage of faculty with PhDs, the student faculty ratio, and whether or not the school is private – all statistics typically associated with school quality. Here is what I found:

1. For every increase of 10 points of the average enrolled student's total SAT score, out-of-state tuition is expected to increase by \$133 dollars
2. If the school employs 10% more faculty members with PhDs, the tuition increases by an average of \$395.80
3. If a student faculty ratio shrinks by 1 unit, the predicted tuition increases by \$68.81

These variables, cumulatively, explain 62.1% of the difference across schools, which is pretty strong. Therefore, now having this model, we can compare the predicted tuition of these schools to their actual tuition data, creating the “bargain score” you were asking for. If a school has a higher, positive bargain score, this means that the school charges less than expected based on its quality. While you can ask me for further information on certain schools you are interested in, the best performers on my raw bargain score metric are Grove City College, Ambassador University, and Lipscomb University. While the tuition model predicted that Grove City College's tuition would be \$13,217, it actually only charges \$5,224.

In case you hope that (sister) wants to stay closer to home, I also created a model using in-state tuition and looked specifically at schools in Ohio. This model produced some interesting results as well.

1. In Ohio, you pay, on average, a higher premium for a heavily accredited faculty base. For a 10% increase in PhDs among faculty, you are expected to pay \$635.20 more.
2. Overall, it seems that prestigious and more expensive schools in Ohio tend to be private schools. In fact private schools charge about \$8,806 more than public schools with otherwise similar characteristics. This is in contrast to \$4,329 nationwide.

Interestingly, the Ohio model claims to account for 84.77% of variability in price among schools in the state, even higher than the nationwide average. Therefore, results should be even more important and accurate. When applying the same bargain calculator to schools in Ohio, an interesting result appeared: private schools dominated the top of the rankings. Tuition to Cedarville College is \$4,152 less than expected by the model, closely followed by the University of Dayton, and John Carroll University. The model thinks that the best value public school is the University of Cincinnati, at a tuition of \$3,234.

Section C – Father

Dad, I know that building my sister's math skills is an important trait for colleges you are interested in. From my dataset, the math SAT score of enrolled students is the closest proxy I have for this trait. Since I know you want her to go to a school outside of Ohio, here is a list of the 5 strongest non-Ohio schools by average SAT math score:

1. MIT (MA) – 742
2. Harvey Mudd College (CA) – 740
3. Princeton (NJ) – 708
4. Dartmouth (NH) – 692
5. Duke (NC) – 68

One thing to keep in mind is that these schools tend to be more expensive. A 10 point increase in average math SAT score among the student population predicts a \$120.80 increase in out-of-state tuition for schools outside of Ohio. Out of the list above, an intriguing school to look at would be Harvey Mudd College, which has the second-highest math SAT, but a positive value in the bargain score model I previously created for mom.

However, these are some of the premier schools in the country. Given that my sister seems the most worried about being accepted into schools, it may prove difficult to convince her into applying to these schools, which likely have low acceptance rates.

Appendix

Creating Key Variables:

```
> # Acceptance rate: lower = more selective
> data$accept_rate <- data$acceptance / data$applications
>
> # Private dummy: 1 = private, 0 = public
> # Original coding: private = 1 for public, private = 2 for private
> data$private_dummy <- ifelse(data$private == 2, 1, 0)
>
> # Total SAT score
> data$total_sat <- data$mathsat + data$verbalsat
```

Section A:

```
> # Average acceptance rate by public/private
> aggregate(accept_rate ~ private_dummy, data = data, mean)
  private_dummy accept_rate
1              0    0.7100500
2              1    0.7567791

> # Simple regression
> model_A1 <- lm(accept_rate ~ private_dummy, data = data)
> summary(model_A1)

Call:
lm(formula = accept_rate ~ private_dummy, data = data)

Residuals:
    Min       1Q   Median       3Q      Max
-0.60229 -0.06688  0.02999  0.09671  0.28995

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.710050   0.009457  75.085 < 2e-16 ***
private_dummy 0.046729   0.011428   4.089 4.82e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1422 on 715 degrees of freedom
Multiple R-squared:  0.02285,    Adjusted R-squared:  0.02149
F-statistic: 16.72 on 1 and 715 DF,  p-value: 4.821e-05

> model_A2 <- lm(accept_rate ~ private_dummy + applications + mathsat + verbalsat, data =
data)
> summary(model_A2)
```

```

Call:
lm(formula = accept_rate ~ private_dummy + applications + mathsat +
    verbalsat, data = data)

Residuals:
    Min       1Q   Median       3Q      Max
-0.52801 -0.06875  0.01734  0.08502  0.33095

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.107e+00  4.015e-02  27.576 < 2e-16 ***
private_dummy  2.981e-02  1.216e-02   2.452  0.0144 *
applications -1.033e-05  1.561e-06 -6.617 7.23e-11 ***
mathsat      1.220e-04  1.967e-04   0.621  0.5351
verbalsat    -9.088e-04  2.251e-04 -4.037 6.01e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1271 on 712 degrees of freedom
Multiple R-squared:  0.2228,    Adjusted R-squared:  0.2184
F-statistic: 51.03 on 4 and 712 DF,  p-value: < 2.2e-16

```

Section B:

```
> model_B <- lm(outofstatetuition ~ total_sat + phdfal + stufalratio + private_dummy, data =
data)
```

```
> summary(model_B)
```

```

Call:
lm(formula = outofstatetuition ~ total_sat + phdfal + stufalratio +
    private_dummy, data = data)

Residuals:
    Min       1Q   Median       3Q      Max
-7993.3 -1629.8   48.8  1560.8 10684.9

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -7666.1343    882.3262  -8.689 < 2e-16 ***
total_sat     13.3287     0.9843   13.541 < 2e-16 ***
phdfal       39.5757     6.7496   5.863 6.94e-09 ***
stufalratio  -68.8113    17.5378  -3.924 9.57e-05 ***
private_dummy 4329.1645    216.5489   19.992 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2421 on 712 degrees of freedom
Multiple R-squared:  0.621,    Adjusted R-squared:  0.6188
F-statistic: 291.6 on 4 and 712 DF,  p-value: < 2.2e-16

```

```
> # Predicted tuition based on quality
```

```
> data$predicted_tuition <- predict(model_B)
```

```
>
```

```
> # Bargain score: positive means school charges less than predicted
```

```
> data$bargain_score <- data$predicted_tuition - data$outofstatetuition
```

```
>
```

```
> # Top 10 bargain schools
```

```
> bargains <- data[order(-data$bargain_score),
```

```
+      c("school", "state", "outofstatetuition", "predicted_tuition",
+       "bargain_score", "total_sat", "phdfal", "stufalratio")
>
> head(bargains, 10)
```

	school	state	outofstatetuition	predicted_tuition	bargain_score	total_sat
222	Grove City College	PA	5224	13217.34	7993.342	1144
9	Ambassador University	TX	2700	10273.69	7573.695	1009
152	David Lipscomb University	TN	5775	12449.25	6674.246	980
717	York College of Pennsylvania	PA	4990	11527.81	6537.810	986
231	Harding University	AR	5504	11850.21	6346.207	1020
413	Piedmont College	GA	5640	11952.54	6312.542	951
96	Christendom College	VA	8730	14805.44	6075.441	1136
639	University of Texas at Dallas	TX	4536	10338.46	5802.462	1188
324	Marshall University	WV	5094	10763.84	5669.842	915
188	Flagler College	FL	5120	10657.55	5537.554	989

	phdfal	stufalratio
222	65	18.4
9	34	17.2
152	97	16.2
717	75	18.1
231	71	17.7
413	89	13.2
96	92	9.3
639	91	20.8
324	77	16.6
188	52	18.1

```
> # Split the data
> ohio_schools <- subset(data, state == "OH")
> non_ohio_schools <- subset(data, state != "OH")
>
> # Model for Ohio schools using in-state tuition
> model_B_ohio <- lm(instatetuition ~ total_sat + phdfal + stufalratio + private_dummy,
+                    data = ohio_schools)
>
> summary(model_B_ohio)
```

```
Call:
lm(formula = instatetuition ~ total_sat + phdfal + stufalratio +
    private_dummy, data = ohio_schools)
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-4152.4 -1254.7   457.1  1066.9  3000.5
```

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  -11823.347    4473.276   -2.643  0.01351 *
total_sat       13.239      4.568     2.898  0.00736 **
phdfal         63.519     28.671     2.215  0.03535 *
stufalratio   -132.901    128.527    -1.034  0.31029
private_dummy  8805.851    1197.743     7.352 6.58e-08 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 2062 on 27 degrees of freedom
Multiple R-squared:  0.8477,    Adjusted R-squared:  0.8252
F-statistic: 37.58 on 4 and 27 DF,  p-value: 1.149e-10
```

```

> # Predicted in-state tuition for Ohio schools
> ohio_schools$predicted_tuition <- predict(model_B_ohio)
>
> # Bargain score: positive means school charges less than predicted
> ohio_schools$bargain_score <- ohio_schools$predicted_tuition - ohio_schools$instatetuition
>
> # Top Ohio bargain schools
> ohio_bargains <- ohio_schools[order(-ohio_schools$bargain_score),
+
+       c("school", "state", "instatetuition",
+       "predicted_tuition", "bargain_score",
+       "total_sat", "phdfal", "stufalratio")]
>
> head(ohio_bargains, 10)

```

	school	state	instatetuition	predicted_tuition	bargain_score	total_sat	phdfal	stufalratio
84	Cedarville College	OH	7344	11496.420	4152.4203	1010	50	15.3
574	University of Dayton	OH	11380	14485.158	3105.1578	1082	81	14.8
272	John Carroll University	OH	11700	14781.637	3081.6372	1063	89	14.5
386	Notre Dame College of Ohio	OH	8064	10795.171	2731.1707	838	61	8.7
49	Bluffton College	OH	9900	12619.371	2719.3706	950	76	13.3
570	University of Cincinnati	OH	3234	5593.748	2359.7483	997	89	10.8
196	Franciscan University of Steubenville	OH	9650	11914.778	2264.7780	1004	57	14.9
714	Xavier University	OH	11520	13037.808	1517.8077	996	73	13.3
27	Baldwin-Wallace College	OH	10995	12161.977	1166.9768	997	68	17.6
112	College of Mount St. Joseph	OH	9800	10609.023	809.0231	920	46	11.1

Section C

```

> non_ohio <- subset(data, state != "OH")
>
> top_math <- non_ohio[order(-non_ohio$mathsat),
+
+       c("school", "state", "mathsat", "verbalsat",
+       "total_sat", "outofstatetuition", "bargain_score")]
>
> head(top_math, 5)

```

	school	state	mathsat	verbalsat	total_sat	outofstatetuition	bargain_score
332	Massachusetts Institute of Technology	MA	742	639	1381	20100	-1807.050
234	Harvey Mudd College	CA	740	630	1370	17230	1086.652
422	Princeton University	NJ	708	647	1355	19900	-2153.223
151	Dartmouth College	NH	692	627	1319	19545	-1865.150
161	Duke University	NC	687	615	1302	18590	-1157.381

```

> model_C <- lm(outofstatetuition ~ mathsat + verbalsat + phdfal + stufalratio + private_dummy,
data = data)
> summary(model_C)

```

```
Call:
lm(formula = outofstatetuition ~ mathsat + verbalsat + phdfal +
    stufalratio + private_dummy, data = data)
```

Residuals:

Min	1Q	Median	3Q	Max
-7971	-1641	53	1565	10747

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-7703.857	889.536	-8.661	< 2e-16	***
mathsat	12.080	3.729	3.239	0.001255	**
verbalsat	14.775	4.283	3.450	0.000594	***
phdfal	39.789	6.782	5.867	6.80e-09	***
stufalratio	-68.726	17.550	-3.916	9.87e-05	***
private_dummy	4309.602	223.893	19.248	< 2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2422 on 711 degrees of freedom
Multiple R-squared: 0.621, Adjusted R-squared: 0.6184
F-statistic: 233 on 5 and 711 DF, p-value: < 2.2e-16